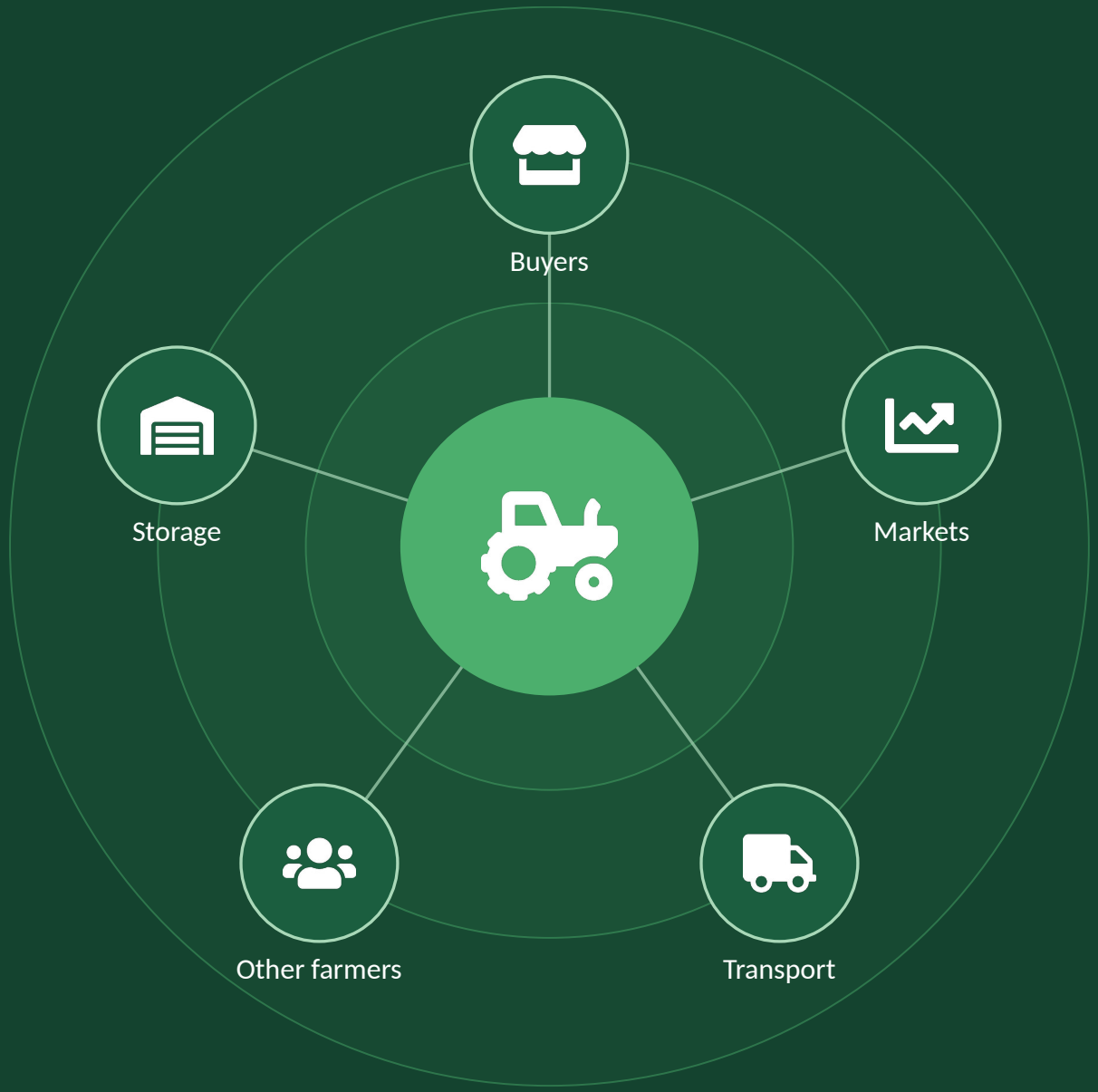
KISANSETU

An Agentic AI-Powered Agricultural Market Access & Intelligent Selling Network

From Harvest to Better Markets

Connecting farmers to market intelligence, buyers, collective bargaining and logistics through coordinated AI agents.



Tech4Bharat 2026 | Agriculture and Food Processing | Problem Statement 6.1

THE PROBLEM

Growing the produce is only half the challenge. Getting a fair return is the other half

Information asymmetry

Net prices across markets, and how they are shifting, are hard to see.

Fragmented supply

Individual quantities are too small to negotiate with larger buyers.

Limited buyer access

Dependence on a narrow set of local buyers and intermediaries.



Small and marginal farmer

Six connected problems. Fixing one alone rarely lifts the final earning.

Transportation economics

A higher price can vanish after transport and handling costs.

Time pressure

Perishable produce leaves little time to search for better options.

Transaction uncertainty

Quantity disputes, quality deductions and delayed payments erode trust.

WHY EXISTING APPROACHES FALL SHORT

Farmers have access to information. What they need is coordinated action

Conventional information tools

- Display market prices
- Show buyer listings
- Provide individual recommendations
- Show transport information
- Display market trends
- Provide transaction records

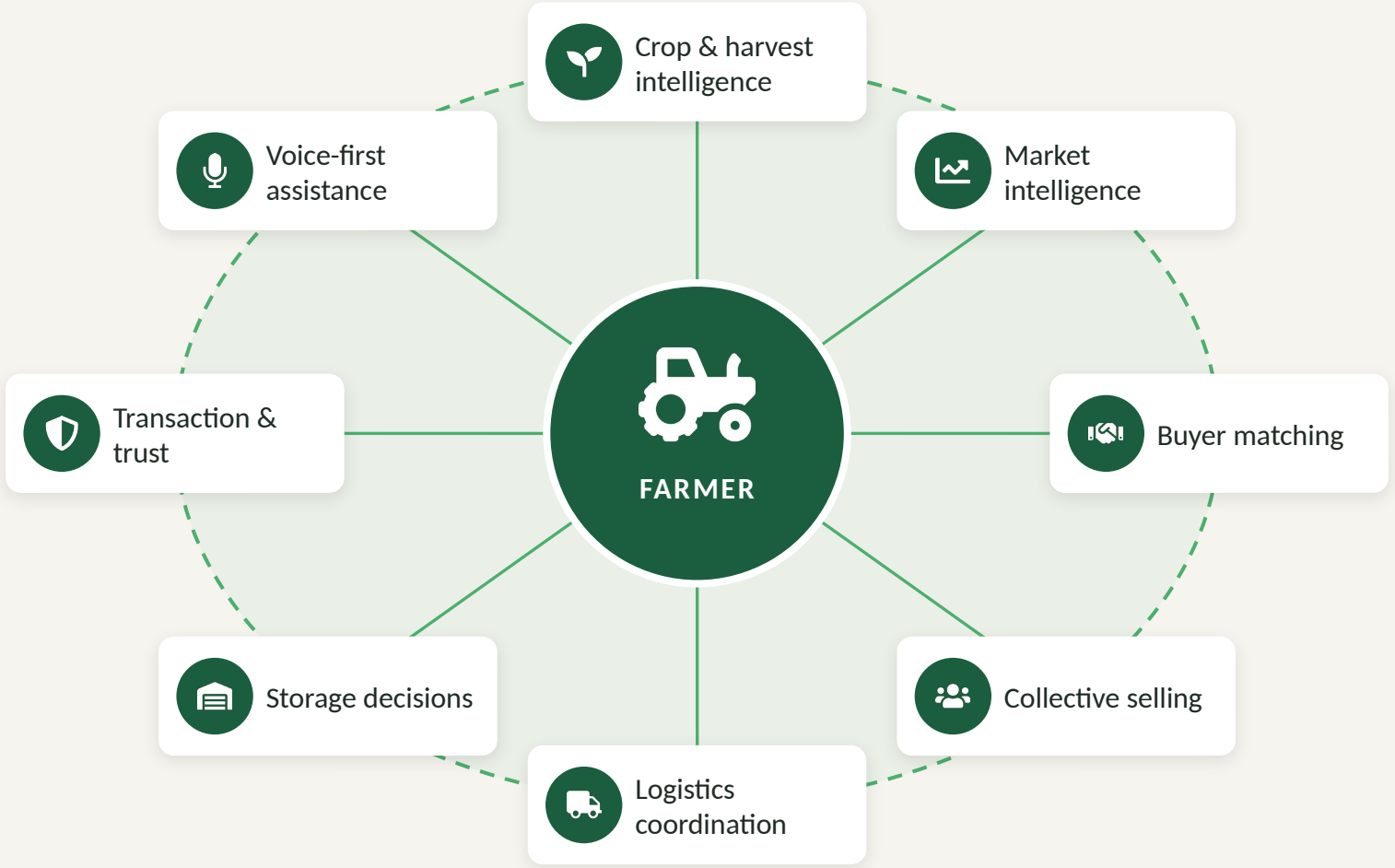
KisanSetu: one shared agentic workflow

- Compare expected net realisation
- Match available supply with buyer demand
- Coordinate collective selling
- Coordinate transport for confirmed sales
- Help evaluate selling opportunities
- Monitor the complete transaction lifecycle

THE SOLUTION

One intelligent network. From anticipated harvest to completed sale

Farmers register upcoming harvests, discover market opportunities, connect with verified buyers, combine produce with other farmers, arrange transport and manage transactions through one coordinated network of AI agents.



Eight services, one workflow

Each service is a step in the same selling journey and shares the same harvest record. None is a separate app.

THE AGENTIC AI ENGINE

Not another chatbot. A network of agents that plans, coordinates and acts

Crop & Harvest Intelligence
Harvest dates, quantities and location, with early local surplus alerts

Market Intelligence
Prices, arrivals, trends, costs and net realisation

Buyer Matching
Matches farmer supply with buyer procurement needs

Collective Selling
Aggregates small lots and evaluates group sales

Central orchestrator
- Understands the farmer's goal
- Maintains shared workflow state
- Delegates tasks to agents
- Evaluates results
- Handles exceptions, replans
- Requests human approval
- Coordinates the next action

Logistics
Vehicles, transport costs, pickup and delivery

Storage Decision
Sell now versus store and sell later

Transaction & Trust
Agreements, delivery records, payments and history

Human approval gate
Financial commitments and consequential decisions need a person

AI agents: intent understanding, planning, tool selection, coordination, exception handling, natural-language interaction

Deterministic algorithms: price calculations, cost estimation, matching constraints, aggregation, optimisation

END-TO-END USE CASE

One farmer. One harvest. A coordinated selling journey

Scenario: a farmer in Gujarat expects to harvest 3,000 kg of tomatoes in five days

ILLUSTRATIVE. Prototype demand is labelled real, sourced or simulated

1 CROP & HARVEST AGENT

Harvest registration

Crop, quantity, location, harvest date and quality, reported by voice.

2 MARKET INTELLIGENCE AGENT

Market investigation

Compares prices, transport, market costs and estimated net realisation.

3 BUYER MATCHING AGENT

Buyer discovery

Finds a food processor needing 5,000 kg within the delivery window.

4 COLLECTIVE SELLING AGENT

Collective supply

Nearby farmers or an FPO supply the other 2,000 kg. Each lot is recorded separately.

5 LOGISTICS AGENT

Logistics planning

Checks vehicle capacity, pickup points and delivery costs.

6 ORCHESTRATOR

Transaction proposal

Price, quantity, deductions, transport cost and estimated proceeds.

7 FARMERS DECIDE

Human approval

Each farmer reviews and approves their own commitment.

8 TRANSACTION & TRUST AGENT

Execution, reconciliation

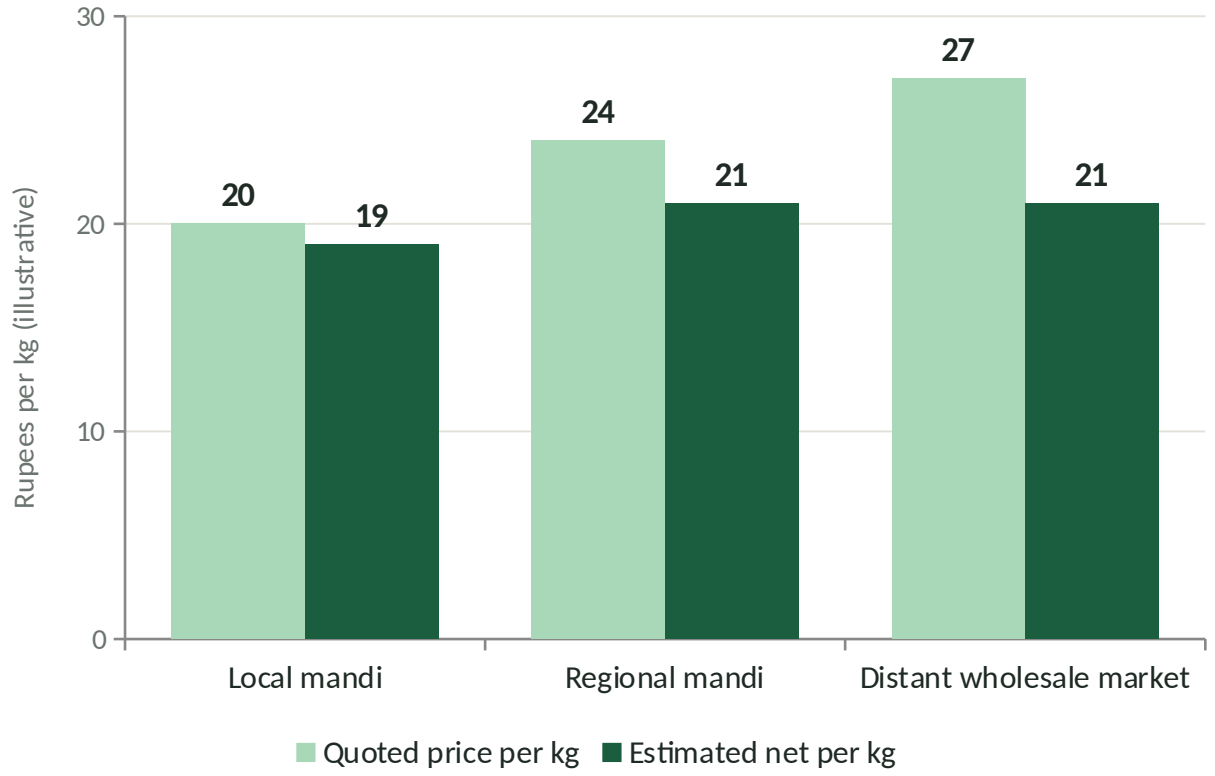
Dispatch, delivery, actual quantities and payment status are tracked.

If conditions change: a buyer withdraws, a transporter drops out or the harvest falls short. The orchestrator replans from the remaining options and the farmers approve again.

WHAT MAKES KISANSETU DIFFERENT

We optimise for what farmers actually earn, not just the price displayed on a screen

Quoted price versus estimated net, per kg — ILLUSTRATIVE FIGURES



Local mandi
Quoted ₹20, costs ₹1 per kg
₹19,000
est. net for 1,000 kg

Regional mandi
Quoted ₹24, costs ₹3 per kg
₹21,000
est. net for 1,000 kg

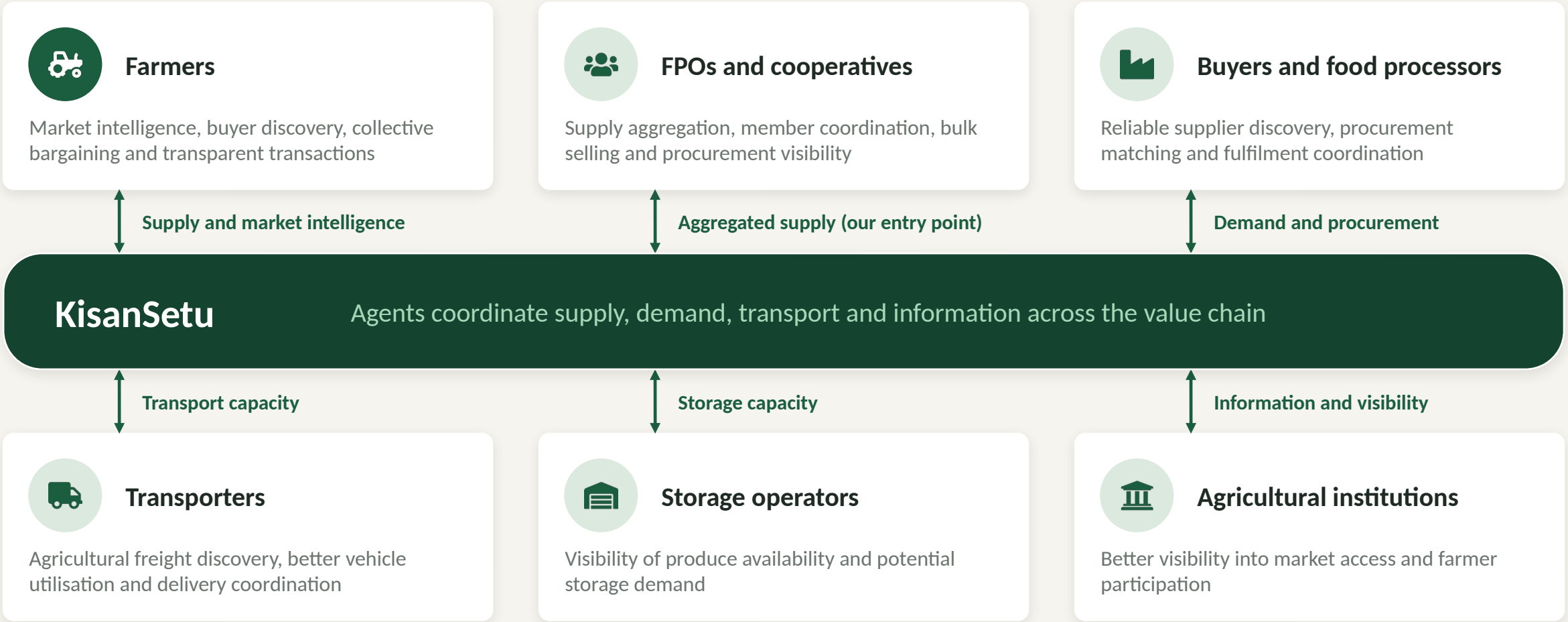
Distant wholesale market
Quoted ₹27, costs ₹6 per kg
₹21,000
est. net for 1,000 kg

The highest price is not the highest earning

The distant market quotes ₹7 per kg more than the local mandi yet nets only ₹2 more, the same as the regional mandi. KisanSetu ranks options by expected net, where cost data is available.

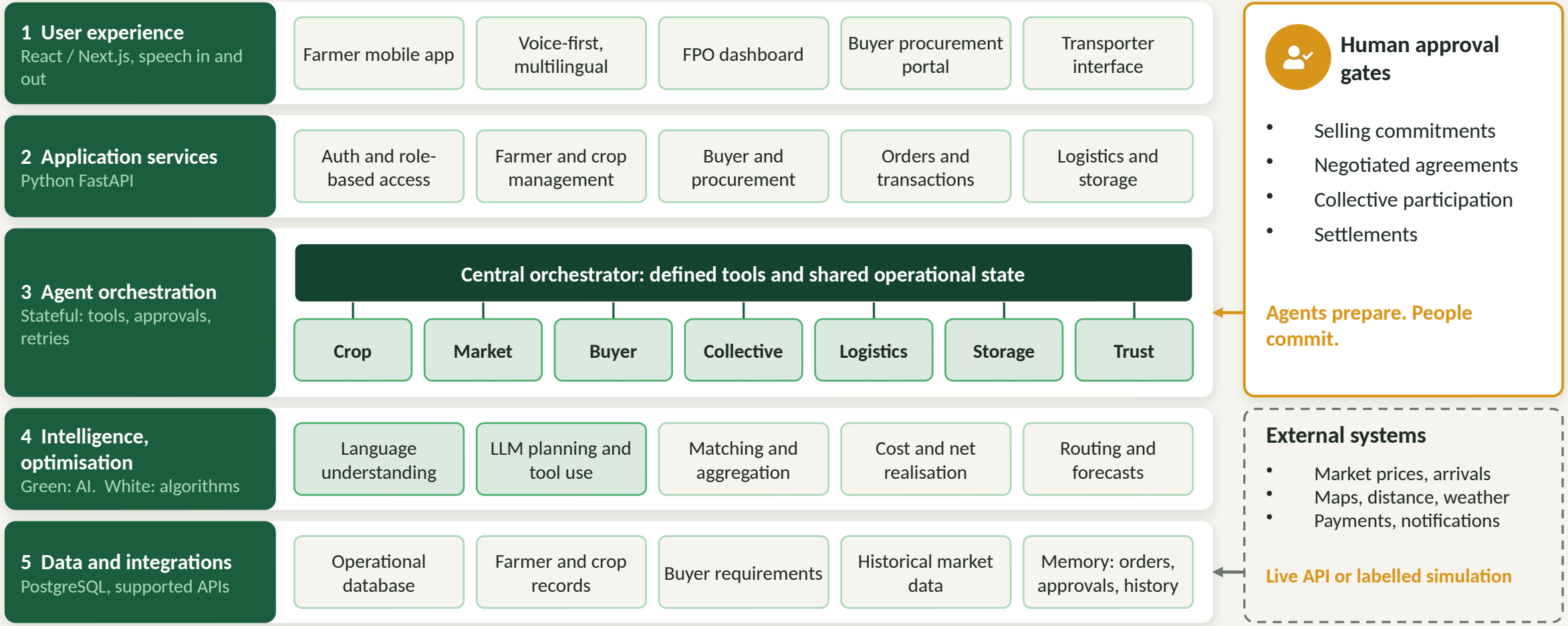
MULTI-STAKEHOLDER ECOSYSTEM

Empowering farmers while connecting the entire agricultural value chain



PROPOSED TECHNICAL ARCHITECTURE

A coordinated agentic architecture built around shared state, reliable data and human control



INTENDED SOCIAL IMPACT

Better market access can translate into stronger farmer economics

Improved net realisation

Compare actual economic returns, not headline prices

Stronger bargaining power

Aggregate supply collectively to reach larger buyers

Reduced avoidable costs

Better transport coordination and selling decisions

Greater market transparency

Clear records of offers, quantities, deductions and payments

Improved market participation

Small farmers reach procurement that is hard to access alone

Impact measurement framework

TARGET METRICS, NOT RESULTS

1. Change in net realisation per kg
2. Farmers accessing new buyers
3. Volume of produce sold collectively
4. Transport cost per kg
5. Share of transactions completed
6. Average payment reconciliation time
7. Farmer repeat usage and retention

A farmer-first platform with multiple institutional revenue opportunities

FPOs and cooperatives

Subscription or licence for aggregation and member coordination

Buyers and processors

Subscription for supplier discovery and procurement management

Logistics providers

Transparent transaction or booking fees where appropriate

NGOs and livelihood bodies

Institutional deployment and programme partnerships

Government and institutions

Licensing or programme deployment, subject to procurement rules

Farmer access: essential market intelligence and basic selling assistance stay free or highly affordable for small farmers

Scalability roadmap

1. One crop, one defined region
2. Validate with farmers and an FPO
3. Onboard local buyers and transporters
4. More crops, neighbouring markets
5. Extend through FPOs and institutions
6. Regional market-access networks

From fragmented agricultural supply to intelligent market access

Every farmer should have access to the market intelligence, buyers, bargaining power and coordination capabilities needed to obtain a fair and informed return for their produce.



KisanSetu: Connecting Every Harvest to Better Opportunities

Tech4Bharat 2026 | Challenge 6.1 | Thank you